NIGHTPULSE

A Low-Cost Stereoscopic Infrared Vision Prototype

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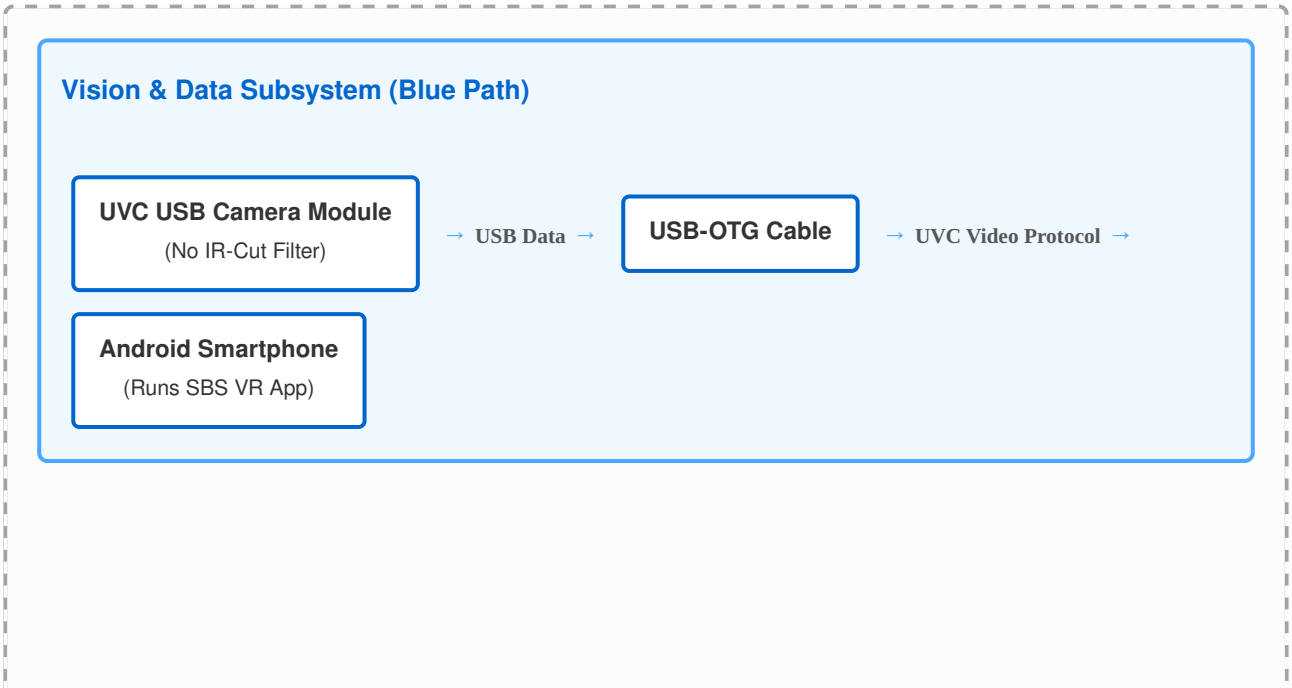
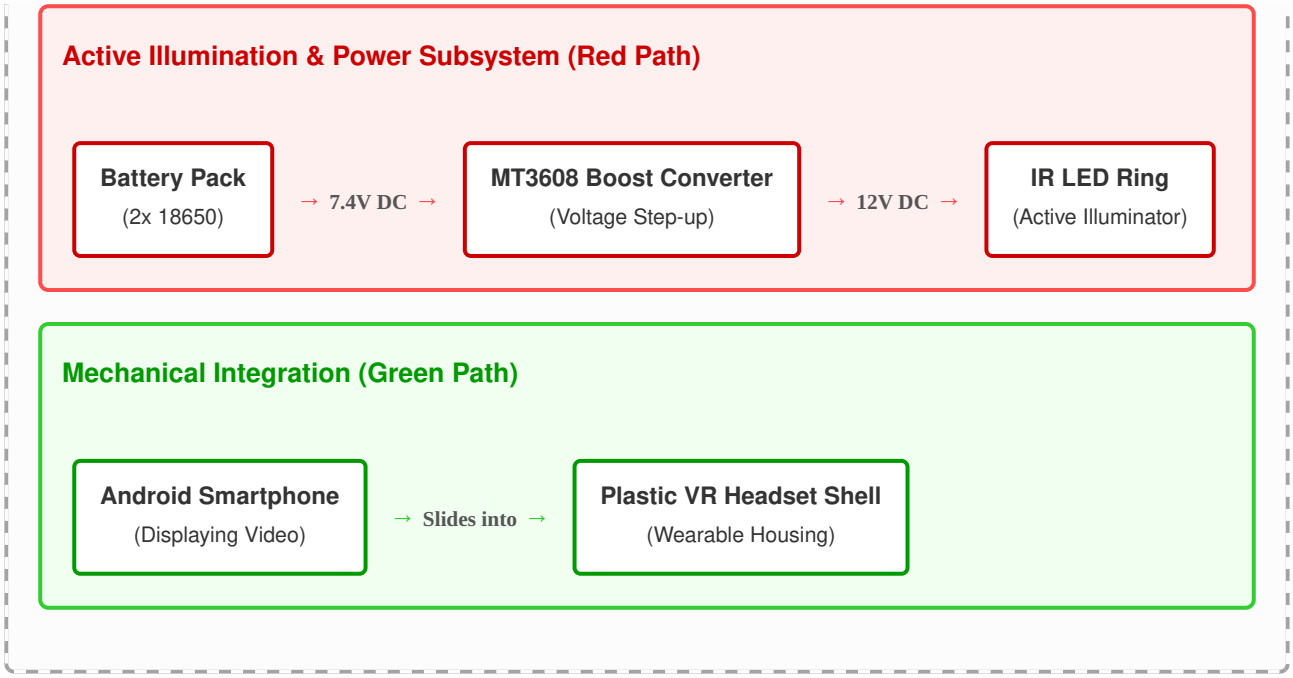
1. Introduction & Methodology

Professional night vision systems are prohibitively expensive for civilian, academic, or low-resource emergency use. Traditional goggles rely on costly analog image intensifier tubes or expensive thermal microbolometers. **NightPulse** bridges this financial gap by proposing a methodology that repurposes off-the-shelf consumer electronics into a cohesive, wearable active-infrared (IR) night vision device.

The core methodology centers on modifying standard digital CMOS cameras by removing their factory-installed IR-cut filters, thereby rendering the image sensors highly sensitive to near-infrared light. To provide illumination in total darkness, an array of 940nm invisible IR LEDs is utilized. The captured video feed is routed via USB OTG to an integrated smartphone. Custom software processes this video feed into a stereoscopic side-by-side (SBS) display format. By housing this system within a standard VR headset shell, the prototype delivers an immersive, hands-free spatial navigation experience.

2. System Architecture & Connection Diagram





3. Proposed Outcome

The primary outcome of the NightPulse project is the successful development of a fully functional, low-cost wearable night vision prototype capable of zero-light navigation. The system is expected to provide clear, high-resolution monochrome visual feedback in pitch-black environments without emitting light visible to the naked human eye.

By effectively democratizing active-IR technology, the project demonstrates that effective spatial navigation in the dark does not require military-grade budgets. The proposed stereoscopic setup aims to maintain user depth perception, reducing disorientation and proving the viability of fused smartphone VR capabilities with modified digital sensors for accessibility and practical civilian use.